Project Objective

The goal of this project is to analyze probability patterns from student data and predict whether a student will pass the final mathematics exam. The dataset contains study hours, attendance, group discussion participation, previous test score and final exam result.

[5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib_venn import venn2
```

[6]:
```
np.random.seed(42)

n = 200

study_hours = np.random.randint(1,20,n)
attendance = np.random.randint(50,100,n)
group_discussion = np.random.choice(["Yes","No"],n)
previous_test_score = np.random.randint(40,100,n)

final_exam_pass = []

for score in previous_test_score:
    if score > 60:
        final_exam_pass.append("Pass")
    else:
        final_exam_pass.append("Fail")

df = pd.DataFrame({
    "study_hours":study_hours,
    "attendance":attendance,
    "group_discussion":group_discussion,
    "previous_test_score":previous_test_score,
    "final_exam_pass":final_exam_pass
})
```

```
df.head()
```

[6]:

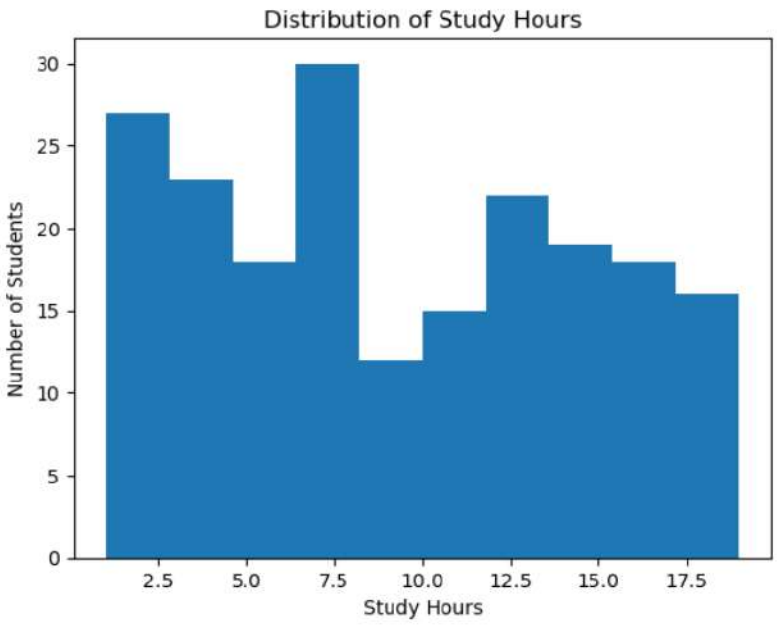
	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
0	7	60	No	44	Fail
1	15	77	Yes	74	Pass
2	11	74	Yes	51	Fail
3	8	99	No	64	Pass
4	7	72	No	91	Pass

[19]:
```
plt.figure()

plt.hist(df['study_hours'], bins=10)

plt.title("Distribution of Study Hours")
plt.xlabel("Study Hours")
plt.ylabel("Number of Students")

plt.show()
```



This histogram shows the distribution of study hours among students. Most students study between 5 and 15 hours per week. Students who study more hours are more

1 Understanding the Basics

What is Probability

Probability is the measure of how likely an event is to occur. It ranges from 0 to 1.

0 means impossible event
1 means certain event

Key Probability Terminology

Event : A specific outcome
Sample Space : All possible outcomes
Probability : Likelihood of an event occurring
Random Variable : Numeric representation of outcomes

Examples of Probability Events from Dataset

1 Student studies more than 10 hours per week
2 Student attendance is greater than 80 percent
3 Student passes the final exam

[7]:
```
pass_count = (df['final_exam_pass']=="Pass").sum()

empirical_probability = pass_count / len(df)

empirical_probability
```

[7]:
```
np.float64(0.65)
```

2 Types of Events

Empirical Probability

Empirical probability is calculated using real data observations.

Theoretical Probability

Theoretical probability is calculated using mathematical formulas based on expected outcomes.

[8]:
```
theoretical_probability = 1/2
theoretical_probability
```

[8]:
```
0.5
```

3 Random Variable and Probability Distribution

Let X be the number of students passing the exam out of 3 randomly selected students.

Possible values of X

0 students pass
1 student passes
2 students pass
3 students pass

[9]:
```
x = np.array([0,1,2,3])
p = np.array([0.05,0.25,0.45,0.25])

distribution = pd.DataFrame({
    "X (Students Passing)":x,
    "Probability":p
})

distribution
```

[9]:

	X (Students Passing)	Probability
0	0	0.05
1	1	0.25
2	2	0.45
3	3	0.25

[10]:
```
mean = np.sum(x*p)

variance = np.sum((x-mean)**2 * p)

print("Mean:",mean)
print("Variance:",variance)
```

```
Mean: 1.9
Variance: 0.69
```

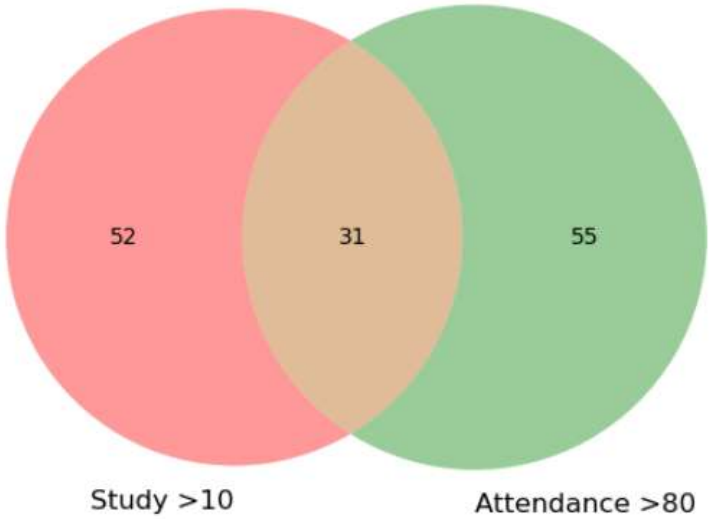
[11]:
```
A = df[df.study_hours > 10]
B = df[df.attendance > 80]

A_count = len(A)
B_count = len(B)

both = len(df[(df.study_hours > 10) & (df.attendance > 80)])
```

```
venn2(subsets = (A_count-both,B_count-both,both),
      set_labels=('Study >10','Attendance >80'))

plt.show()
```



[12]:
```
table = pd.crosstab(df['group_discussion'],df['final_exam_pass'])

table
```

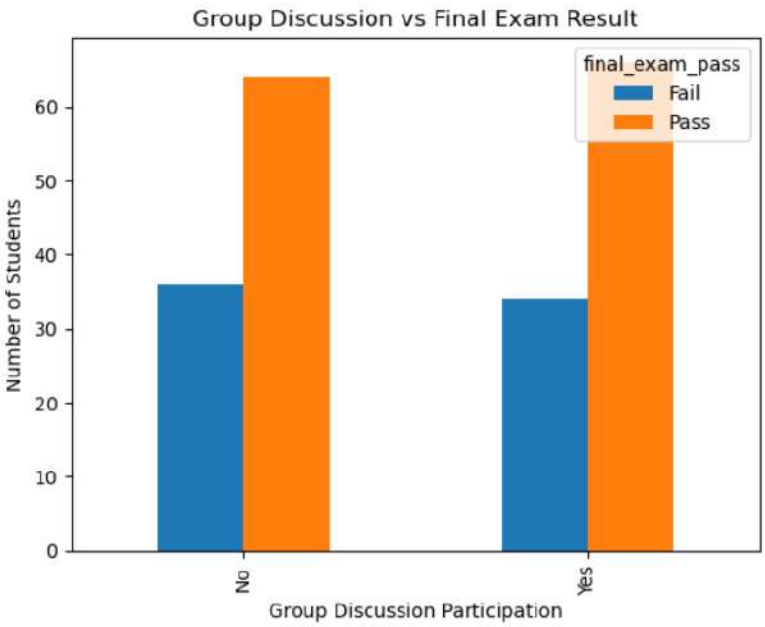
[12]:

final_exam_pass	Fail	Pass
group_discussion		
No	36	64
Yes	34	66

[21]:
```
table.plot(kind='bar')

plt.title("Group Discussion vs Final Exam Result")
plt.xlabel("Group Discussion Participation")
plt.ylabel("Number of Students")

plt.show()
```



The chart shows the relationship between group discussion participation and exam results. Students who participate in discussions tend to have higher pass rates compared to those who do not.

[13]:
```
joint_probability = len(df[(df.group_discussion=="Yes") & (df.final_exam_pass=="Pass")]) / len(df)

joint_probability
```

[13]:
```
0.33
```

[14]:
```
marginal_probability = pass_count / len(df)

marginal_probability
```

[14]:
```
np.float64(0.65)
```

[15]:
```
conditional_probability = table.loc["Yes","Pass"] / table.loc["Yes"].sum()

conditional_probability
```

[15]:
```
np.float64(0.66)
```

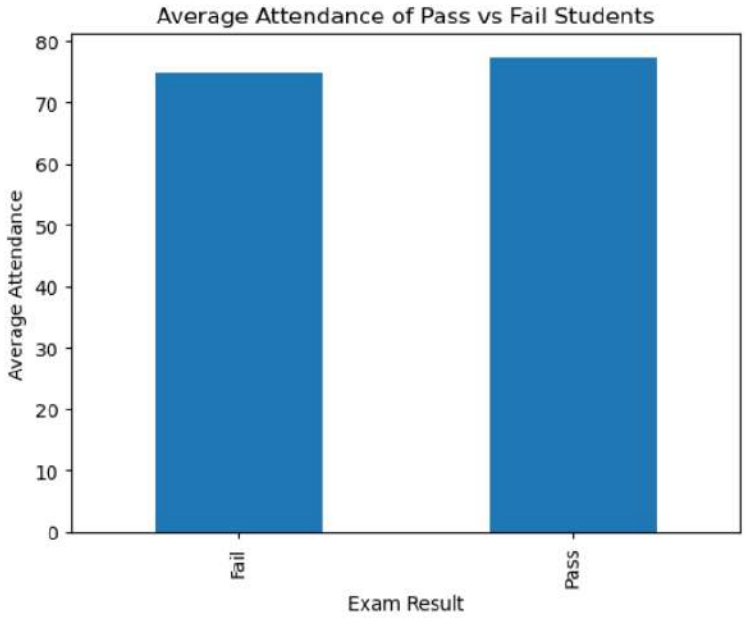
[20]:
```
attendance_pass = df.groupby('final_exam_pass')['attendance'].mean()

plt.figure()

attendance_pass.plot(kind='bar')

plt.title("Average Attendance of Pass vs Fail Students")
plt.xlabel("Exam Result")
plt.ylabel("Average Attendance")

plt.show()
```



This chart shows the average attendance of students who passed and failed the exam. Students who passed generally have higher attendance compared to those who failed.

6 Understanding Relationships

Conditional probability shows how the probability of passing changes when students participate in group discussions.

If students who participate in discussions have higher probability of passing the exam, then the events are dependent.

This means participation in group discussions increases the chance of passing the exam.

7 Bayes Theorem Application

Given:

70 percent of students who pass had high attendance
40 percent of students who fail also had high attendance
60 percent of all students had high attendance

Using Bayes theorem we can calculate the probability that a student passed the exam given that they had high attendance.

[17]:
```
P_A = 0.6
P_B_given_A = 0.7
P_B = 0.6

bayes = (P_B_given_A * P_A) / P_B

bayes
```

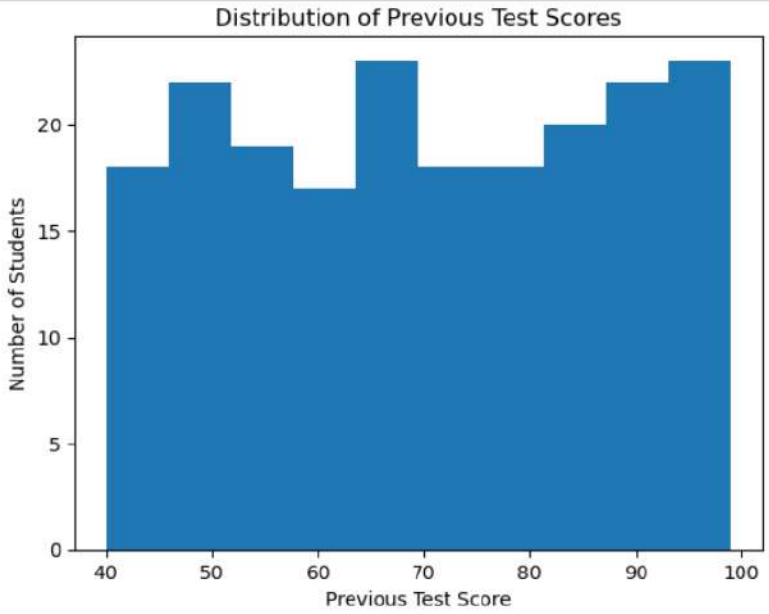
[17]:
```
0.7
```

[22]:
```
plt.figure()

plt.hist(df['previous_test_score'], bins=10)

plt.title("Distribution of Previous Test Scores")
plt.xlabel("Previous Test Score")
plt.ylabel("Number of Students")

plt.show()
```



This histogram shows the distribution of previous test scores. Students with higher previous test scores tend to have a higher probability of passing the final exam.

Final Summary ¶

The analysis shows that students who study more hours, attend classes regularly and participate in group discussions have a higher probability of passing the exam.

Attendance above 80 percent and higher previous test scores are strong indicators of final exam success.

Therefore study habits, attendance and participation significantly affect the probability of passing the exam.